

ALFACAREERS ENTERPRISE AI RECRUITMENT PLATFORM

System Architecture, Multi-Portal Progress Report & Strategic Vision Document

1. Executive Vision & Platform Overview

AlfaCareers is a multi-tenant Enterprise SaaS B2B2C AI Recruitment Marketplace designed to bridge the gap between high-caliber candidates, enterprise employers, and platform governance. Built on a modern decoupled stack using **FastAPI (Python 3.14)**, **Next.js 14 (TypeScript)**, and **ChromaDB Vector Store**, AlfaCareers replaces traditional manual screening with dynamic cosine similarity vector matching, automated HR email dispatches, and an AI-driven recruitment pipeline.

Target Market MENA & Global Tech Hubs (Dubai, Abu Dhabi, Lahore)	Core Architecture 3 Independent Portals + Dual Vector Stores	System Quality 100% Verified E2E Integration Suite
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2. Completed Multi-Portal Feature Inventory

Portal Scope	Route Path	Implemented Core Modules & Telemetry	Status
Candidate Co-Pilot Portal	/dashboard	- Multi-CV Upload & Master CV Selector - Structured Salary & Preferences Matrix - Dynamic Total Experience Engine - Llama 3.1 AI Executive Summary Generator - ChromaDB Cosine Vector Matched Job Feed - ReportLab Single-Page ATS Resume PDF Compiler - Application History with AI Skill Gap Telemetry	100% COMPLETE
Employer Enterprise Portal	/employer	- Enterprise Company Onboarding & Branding - AI Job Post Creator with Vector Embedding Tag Manager - Active Jobs Directory with Live Applicant Counters 5-Stage Applicant Screening Kanban Board - Candidate AI Match Analysis Modal & PDF Exporter	100% COMPLETE
Super Admin Governance Center	/admin	- Real-Time Executive Analytics Overview - Employer Verification Queue (Approve / Revoke) - Light Corporate Paginated Job Moderation Queue - Vector Store Indexing Trigger Control - Searchable Multi-Tenant User Governance Directory	100% COMPLETE

3. AI & Vector Matching Architecture

AlfaCareers leverages a dual-collection **ChromaDB Vector Store** architecture operating alongside a local **Llama 3.1 LLM engine (via Ollama API)** to deliver intelligent semantic matching and summary generation:

Component	Technical Specifications & Execution Role
ChromaDB Dual Store	Indexes candidates_vector_store and jobs_vector_store using sentence-transformer embeddings to perform cosine similarity calculations yielding match scores up to 99.2%.
Llama 3.1 Local LLM	Generates structured 3-bullet AI Executive Summaries for candidate profiles and synthesizes job-to-candidate skill gap explanations.
ATS Resume Compiler	Python ReportLab Flowable engine creating standardized single-page ATS PDF resumes directly from candidate profile JSON representations.

4. Verified Test Suite & System Quality

The entire platform lifecycle is verified by an automated 5-step End-to-End Integration Suite (test_e2e_flow.py) validating multi-role RBAC security, job vector indexing, candidate matching, and live stage sync:

Step & Scope	Trigger Endpoint / Action	Verified Result
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Step 1: Employer Posting	POST /api/v1/employer/jobs	Job created in DB with pending_approval status.
Step 2: Admin Moderation	PUT /api/v1/admin/jobs/{id}/moderate	Approved to published & indexed in ChromaDB.
Step 3: Vector Match & Apply	GET /api/v1/jobs/matched	Candidate gets 92.2% Vector Match & submits application.
Step 4: Kanban Stage Migration	PUT /api/v1/employer/applicants/{id}/stage	Employer advances candidate stage to interview.
Step 5: History Live Sync	GET /api/v1/applications	Candidate history sync confirms status reads interview.

5. Strategic Product Roadmap (Upcoming Phases)

Sprint Phase	Target Capability & Strategic Impact	Priority
Phase 5	Resend / SendGrid Live Email Delivery: Connect transactional email API for live candidate and employer email notifications.	HIGH
Phase 6	Candidate-Employer AI Interview Scheduler: Automated calendar slot booking for shortlisted candidates in Kanban board.	MEDIUM
Phase 7	Automated Deep Web Hunter Cron Scheduler: Recurring background scraper polling target corporate job boards every 6 hours.	MEDIUM